

NANYANG TECHNOLOGICAL UNIVERSITY**SEMESTER 2 EXAMINATION 2024-2025****EE6222 – MACHINE VISION**

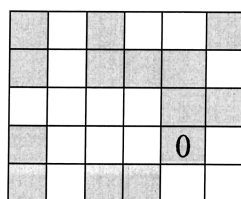
April / May 2025

Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 3 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed-book examination.
5. Unless specifically stated, all symbols have their usual meanings.

1. A binary image is represented by set A .
 - (a) Design an algorithm to extract an 8-connected component from a binary image represented by set A .
(7 Marks)
 - (b) Design an algorithm to extract a 4-connected component from a binary image represented by set A .
(4 Marks)
 - (c) Let set A contain the shaded pixels in Figure 1. The pixel indicated by “0” in Figure 1 is selected to extract an 8-connected component. Write the value n in the proper pixel cells/locations to indicate the sequence of which the pixels are extracted in the n^{th} iteration of your algorithm.

**Figure 1****(7 Marks)**

Note: Question No. 1 continues on page 2.

- (d) Repeat part (c) but show the result for extracting a 4-connected component instead of 8-connected component.

(7 Marks)

2. An image contains 40 fishes of type ω_1 and 60 fishes of type ω_2 . You need to identify the type of each fish in the image. An image segmentation algorithm is used to separate each fish and compute its size. The two PDFs of the size x of the two types of fishes, $p(x|\omega_1)$ and $p(x|\omega_2)$, are shown in Figure 2, where 2 straight lines can be observed.

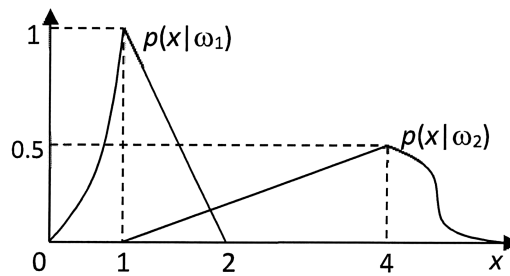


Figure 2

- (a) Determine the type of a fish with $x=1.5$, which minimizes the probability of your wrong decision, and calculate the error rate.

(12 Marks)

- (b) Generate a decision rule to determine the type of fish based on x , which minimizes the probability of your wrong decision, and evaluate the error rate of your method.

(13 Marks)

3. A convolution layer of a two-dimensional CNN generates q output feature maps from p input feature maps by performing 1×1 convolution. Each input pixel is denoted by x_{ijk} , $1 \leq i \leq m$, $1 \leq j \leq n$, $1 \leq k \leq p$, and each output pixel is denoted by y_{ijk} .

- (a) Express the outputs in terms of inputs through the learnable filter parameters.

(8 Marks)

- (b) Determine the numbers of the inputs, outputs and learnable parameters, respectively.

(4 Marks)

- (c) Each 2-dimensional feature map is flattened into a sequence of pixels denoted by $\tilde{x}_{lk}$, $1 \leq l \leq mn$ and $\tilde{y}_{lk}$. Express the outputs in terms of inputs through the learnable filter parameters.

(5 Marks)

Note: Question No. 3 continues on page 3.

- (d) Let us represent different pixels by different rows of a matrix, where we use $\mathbf{X}$ for inputs and $\mathbf{Y}$ for outputs, respectively. Express the output matrix $\mathbf{Y}$ in terms of input matrix $\mathbf{X}$ through the learnable filter parameter matrix $\mathbf{W}$.

(8 Marks)

4. Deep learning has been used for video processing and stereo matching.

- (a) Generative models are a class of machine learning models designed to generate new data samples that resemble a given dataset. Answer the following questions:

- (i) Explain the key difference between discriminative models and generative models.
- (ii) Describe how a Variational Autoencoder (VAE) works. What is the role of the latent space in a VAE?
- (iii) Compare the Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) in terms of their architecture, training process, and applications.

(10 Marks)

- (b) Stereo matching is a fundamental problem in computer vision that involves finding corresponding points between two images taken from slightly different viewpoints to estimate depth.

- (i) Explain the concept of disparity in stereo matching. How is it related to depth estimation?
- (ii) Describe the key steps involved in a typical stereo matching algorithm.
- (iii) Discuss three major challenges in stereo matching.

(15 Marks)

END OF PAPER

EE6222 MACHINE VISION

Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.